IGCSE 0607 Practice Questions — E7.6

Pythagoras' theorem and trigonometry in 3D

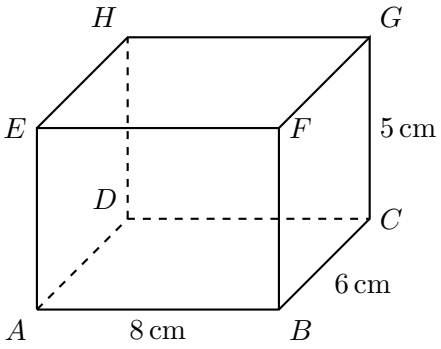
Total Questions: 9 — **Total Marks:** 69 Marks

Table of Contents & Quick Navigation

- **E7.6** Pythagoras' theorem and trigonometry in 3D
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*

---

1 The diagram shows a cuboid $ABCDEFGH$.
$AB = 8$ cm, $BC = 6$ cm and $CG = 5$ cm.



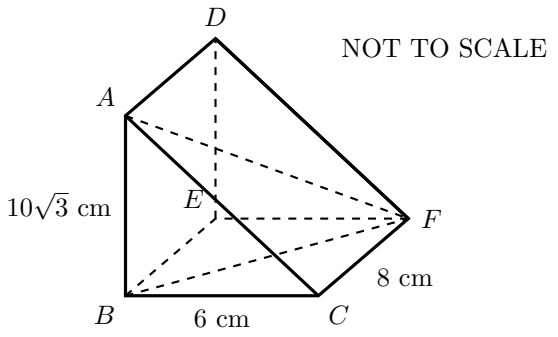
(a) Calculate the exact length of the diagonal AC.

$AC =$ ..... cm **[2]**

(b) Calculate the exact length of the space diagonal AG.
Give your answer in the form $a\sqrt{b}$, where a and b are integers.

$AG =$ ..... cm **[3]**

2 The diagram shows a triangular prism $ABCDEF$ resting on a horizontal rectangular base $BCFE$, where $BC = 6$ cm and $BE = 8$ cm. The rectangular face $ABED$ is vertical with $AB = 10\sqrt{3}$ cm.



(a) Calculate the length of BF.

$BF =$ ..... cm **[2]**

(b) Calculate the length of AF.

$AF =$ ..... cm **[2]**

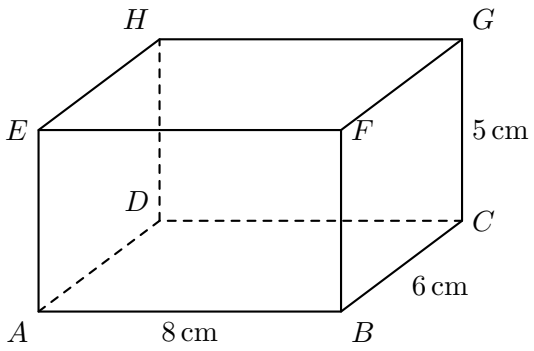
(c) Calculate the size of the angle between the line AF and the horizontal base $BCFE$.

..... ° **[2]**

(d) Calculate the exact area of the rectangular face $ACFD$, giving your answer in the form $k\sqrt{21}$, where k is an integer.

Area = ... cm^2 **[3]**

3 The diagram shows a cuboid $ABCDEFGH$.
$AB = 8$ cm, $BC = 6$ cm and $CG = 5$ cm.



(a) Calculate the length of the diagonal AC.

..... cm **[2]**

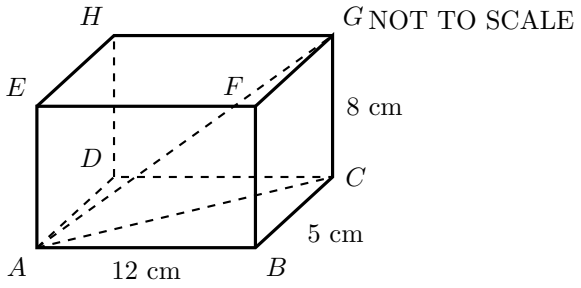
(b) Calculate the length of the space diagonal AG.

..... cm **[2]**

(c) Calculate the angle between the line AG and the base $ABCD$.

..... ° **[2]**

4 The diagram shows a cuboid $ABCDEFGH$ with a horizontal rectangular base $ABCD$.
$AB = 12$ cm, $BC = 5$ cm, and $CG = 8$ cm.



(a) Calculate the length of AC.

$AC =$ ..... cm **[2]**

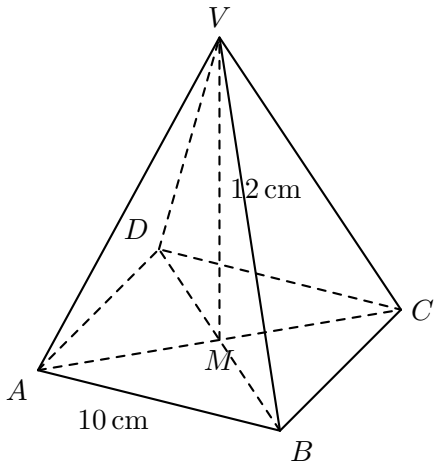
(b) Calculate the length of the diagonal AG, giving your answer correct to 3 significant figures.

$AG =$ ..... cm **[2]**

(c) Calculate the angle that the diagonal AG makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place.

..... ° **[3]**

5 The diagram shows a pyramid $VABCD$ with a square base $ABCD$ of side 10 cm.
The point M is the centre of the base and V is vertically above M.
The vertical height VM of the pyramid is 12 cm.



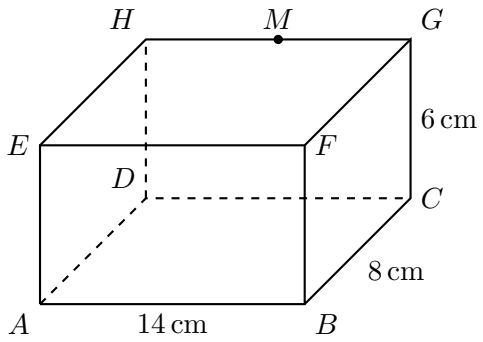
(a) Calculate the length of the edge VA.

..... cm **[3]**

(b) Calculate the angle between the edge VA and the base $ABCD$.

..... ° **[2]**

6 The diagram shows a cuboid $ABCDEFGH$.
$AB = 14$ cm, $BC = 8$ cm and $CG = 6$ cm.
The point M is the midpoint of the edge GH.



(a) Calculate the length of the diagonal BD.

$BD =$ ..... cm **[2]**

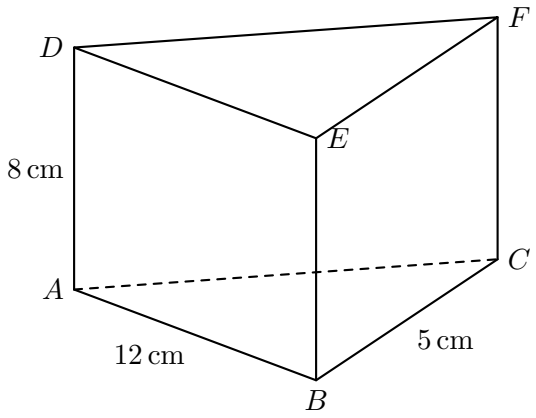
(b) Calculate the angle between the line DF and the base $ABCD$.

Angle = ... ° **[3]**

(c) Calculate the length of AM.

$AM =$ ..... cm **[3]**

7 The diagram shows a right triangular prism $ABCDEF$.
The base ABC is a horizontal right-angled triangle with angle $ABC = 90°$.
$AB = 12$ cm, $BC = 5$ cm and the vertical height $AD = 8$ cm.



(a) Calculate the length of AF.

..... cm **[3]**

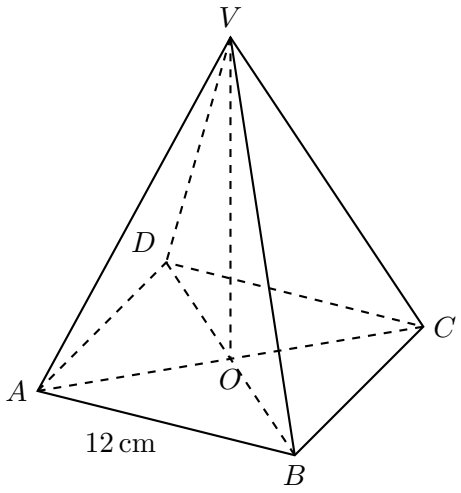
(b) Calculate the angle between the line AF and the base ABC.

..... ° **[2]**

(c) Calculate the angle between the line AF and the vertical plane $ABED$.

..... ° **[3]**

8 The diagram shows a solid right pyramid $VABCD$ with a square base $ABCD$.
The vertex V is directly above the centre O of the base.
The sides of the square base are 12 cm and the volume of the pyramid is 384 cm^3.



(a) Show that the vertical height, VO, is 8 cm.

[2]

(b) Calculate the length of the slant edge VA.

$VA =$ ..... cm **[3]**

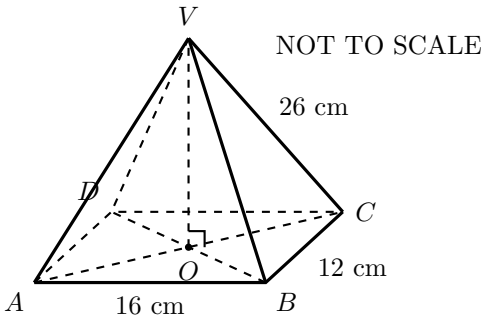
(c) Calculate the angle between the edge VA and the base $ABCD$.

Angle = ... ° **[2]**

(d) Calculate the angle between the triangular face VAB and the base $ABCD$.

Angle = ... ° **[3]**

9 The diagram shows a solid right pyramid $VABCD$ with a horizontal rectangular base $ABCD$ of length $AB = 16$ cm and width $BC = 12$ cm. The apex V is vertically above the centre O of the base, and each slant edge $VA = VB = VC = VD = 26$ cm.



(a) Show that the vertical height VO of the pyramid is 24 cm.

[3]

(b) Calculate the angle that the slant edge VA makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place.

..... ° **[2]**

(c) M is the midpoint of BC.

Calculate

(i) the angle that the triangular face VBC makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place,

..... ° **[3]**

(ii) the shortest distance from the centre O to the triangular face VBC, giving your answer correct to 3 significant figures.

Distance = ... cm **[3]**